

Geometric Transformations in 2D

Translation, Scaling, Rotation, Affine and Projective Transformations

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1 Introduction

Geometric transformations are fundamental to computer vision. They describe how the coordinates of points change when an image is translated, scaled, rotated, sheared, or viewed from a different perspective.

A useful way to represent these transformations is using matrices.

For example, a transformation can be written as

$$\mathbf{p}' = H\mathbf{p}. \quad (1)$$

The important question is:

Where do the individual elements of the transformation matrix come from?

Rather than memorizing transformation matrices, we can derive them directly from the desired equations for the transformed coordinates.

2 Homogeneous Coordinates

A point in a two-dimensional Cartesian coordinate system is

$$\mathbf{p} = \begin{bmatrix} x \\ y \end{bmatrix}. \quad (2)$$

We introduce a third coordinate and represent the same point using homogeneous coordinates:

$$\mathbf{p}_h = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (3)$$

More generally,

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \sim \begin{bmatrix} \lambda x \\ \lambda y \\ \lambda \end{bmatrix}, \quad \lambda \neq 0. \quad (4)$$

The Cartesian coordinates are obtained by dividing by the third coordinate.

Thus,

$$(x, y, w) \longrightarrow \left(\frac{x}{w}, \frac{y}{w} \right). \quad (5)$$

The use of homogeneous coordinates allows translation to be represented using matrix multiplication.

3 Translation

Suppose a point is translated by (t_x, t_y) .

The desired transformation is

$$x' = x + t_x \quad (6)$$

and

$$y' = y + t_y. \quad (7)$$

We want a matrix T such that

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = T \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (8)$$

Consider

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}. \quad (9)$$

Multiplication gives

$$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}. \quad (10)$$

Therefore,

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad (11)$$

The last column contains the translation parameters.

4 Scaling

Suppose a point is scaled by s_x in the x direction and s_y in the y direction.

Then

$$x' = s_x x \quad (12)$$

and

$$y' = s_y y. \quad (13)$$

Therefore,

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (14)$$

Hence,

$$S = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (15)$$

If

$$s_x = s_y = s,$$

the transformation is uniform scaling.

If

$$s_x \neq s_y,$$

the transformation is non-uniform scaling.

Negative values of s_x or s_y produce reflection in the corresponding direction.

5 Rotation

Rotation is slightly more interesting because the elements of the rotation matrix arise from the trigonometric angle addition identities.

Consider a point

$$(x, y) \tag{16}$$

at a distance r from the origin.

Represent the point in polar coordinates:

$$x = r \cos \phi, \quad y = r \sin \phi. \tag{17}$$

Suppose the point is rotated counter-clockwise by θ .

The new coordinates are

$$x' = r \cos(\phi + \theta) \tag{18}$$

and

$$y' = r \sin(\phi + \theta). \tag{19}$$

Using

$$\cos(\phi + \theta) = \cos \phi \cos \theta - \sin \phi \sin \theta \tag{20}$$

and

$$\sin(\phi + \theta) = \sin \phi \cos \theta + \cos \phi \sin \theta, \tag{21}$$

we obtain

$$x' = r \cos \phi \cos \theta - r \sin \phi \sin \theta \tag{22}$$

$$= x \cos \theta - y \sin \theta, \tag{23}$$

and

$$y' = r \sin \phi \cos \theta + r \cos \phi \sin \theta \tag{24}$$

$$= x \sin \theta + y \cos \theta. \tag{25}$$

Therefore,

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{26}$$

6 Composition of Transformations

Multiple transformations can be combined by multiplying their transformation matrices.

For example, suppose we first scale a point, then rotate it, and finally translate it.

Then

$$\mathbf{p}' = TRS\mathbf{p}. \quad (27)$$

The rightmost transformation is applied first.

Thus,

$$\mathbf{p} \xrightarrow{S} S\mathbf{p} \xrightarrow{R} RS\mathbf{p} \xrightarrow{T} TRS\mathbf{p}. \quad (28)$$

An important property is

$$\boxed{AB \neq BA} \quad (29)$$

in general.

Therefore, the order in which geometric transformations are performed matters.

7 Affine Transformation

An affine transformation is represented using the matrix

$$A = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \quad (30)$$

which gives

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (31)$$

Therefore,

$$x' = ax + by + t_x \quad (32)$$

and

$$y' = cx + dy + t_y. \quad (33)$$

There are six unknown parameters:

$$\boxed{a, b, c, d, t_x, t_y} \quad (34)$$

The next question is:

How many point correspondences are required to estimate these parameters?

8 Estimating an Affine Transformation from Point Correspondences

Suppose we know three corresponding point pairs:

$$(x_1, y_1) \longrightarrow (x'_1, y'_1), \quad (35)$$

$$(x_2, y_2) \longrightarrow (x'_2, y'_2), \quad (36)$$

and

$$(x_3, y_3) \longrightarrow (x'_3, y'_3). \quad (37)$$

Each point correspondence gives two equations.

For the first point,

$$x'_1 = ax_1 + by_1 + t_x, \quad (38)$$

$$y'_1 = cx_1 + dy_1 + t_y. \quad (39)$$

For the second point,

$$x'_2 = ax_2 + by_2 + t_x, \quad (40)$$

$$y'_2 = cx_2 + dy_2 + t_y. \quad (41)$$

For the third point,

$$x'_3 = ax_3 + by_3 + t_x, \quad (42)$$

$$y'_3 = cx_3 + dy_3 + t_y. \quad (43)$$

Thus,

$$3 \times 2 = 6 \quad (44)$$

equations are obtained.

Since the affine transformation has six unknowns,

3 point correspondences are sufficient.

(45)

8.1 Solving for the First Row

Consider only the equations involving x' :

$$x'_1 = ax_1 + by_1 + t_x, \quad (46)$$

$$x'_2 = ax_2 + by_2 + t_x, \quad (47)$$

$$x'_3 = ax_3 + by_3 + t_x. \quad (48)$$

These can be written as

$$\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ t_x \end{bmatrix} = \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix}. \quad (49)$$

Define

$$M = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}. \quad (50)$$

Then

$$M \begin{bmatrix} a \\ b \\ t_x \end{bmatrix} = \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix}. \quad (51)$$

If M is invertible,

$$\boxed{\begin{bmatrix} a \\ b \\ t_x \end{bmatrix} = M^{-1} \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix}} \quad (52)$$

8.2 Solving for the Second Row

Similarly, the equations involving y' are

$$y'_1 = cx_1 + dy_1 + t_y, \quad (53)$$

$$y'_2 = cx_2 + dy_2 + t_y, \quad (54)$$

$$y'_3 = cx_3 + dy_3 + t_y. \quad (55)$$

Therefore,

$$M \begin{bmatrix} c \\ d \\ t_y \end{bmatrix} = \begin{bmatrix} y'_1 \\ y'_2 \\ y'_3 \end{bmatrix}. \quad (56)$$

Hence,

$$\boxed{\begin{bmatrix} c \\ d \\ t_y \end{bmatrix} = M^{-1} \begin{bmatrix} y'_1 \\ y'_2 \\ y'_3 \end{bmatrix}} \quad (57)$$

The complete affine transformation is consequently

$$\boxed{A = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix}}. \quad (58)$$

8.3 Geometric Condition

The three source points cannot be collinear.

The matrix

$$M = \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \quad (59)$$

must satisfy

$$\boxed{\det(M) \neq 0}. \quad (60)$$

If the three points are collinear, then

$$\det(M) = 0, \quad (61)$$

and the transformation cannot be uniquely determined.

9 Projective Transformation

An affine transformation cannot model general perspective distortion.

For example, when a planar rectangular object is viewed obliquely, its image may appear as a general quadrilateral.

A projective transformation, also called a homography, is represented by

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad (62)$$

and

$$\begin{bmatrix} x'_h \\ y'_h \\ w' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}. \quad (63)$$

Therefore,

$$x'_h = h_{11}x + h_{12}y + h_{13}, \quad (64)$$

$$y'_h = h_{21}x + h_{22}y + h_{23}, \quad (65)$$

$$w' = h_{31}x + h_{32}y + h_{33}. \quad (66)$$

The Cartesian coordinates are obtained by dividing by w' :

$$\boxed{x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}} \quad (67)$$

and

$$\boxed{y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}}. \quad (68)$$

10 Why Four Correspondences?

The homography matrix has nine elements:

$$h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}, h_{33}. \quad (69)$$

However, a homography is defined only up to an arbitrary non-zero scale.

That is,

$$H \sim \lambda H, \quad \lambda \neq 0. \quad (70)$$

Therefore, only

$$9 - 1 = \boxed{8} \quad (71)$$

independent parameters need to be estimated.

Each point correspondence provides two independent constraints.

Hence,

$$4 \times 2 = 8. \quad (72)$$

Therefore,

$$\boxed{\text{Four point correspondences are sufficient to estimate a homography.}} \quad (73)$$

11 Deriving the Homography Equations

Consider one point correspondence

$$(x_i, y_i) \longrightarrow (x'_i, y'_i). \quad (74)$$

From the projective transformation,

$$x'_i = \frac{h_{11}x_i + h_{12}y_i + h_{13}}{h_{31}x_i + h_{32}y_i + h_{33}}. \quad (75)$$

Cross multiplying,

$$x'_i(h_{31}x_i + h_{32}y_i + h_{33}) = h_{11}x_i + h_{12}y_i + h_{13}. \quad (76)$$

Rearranging,

$$\boxed{h_{11}x_i + h_{12}y_i + h_{13} - h_{31}x_ix'_i - h_{32}y_ix'_i - h_{33}x'_i = 0} \quad (77)$$

Similarly, using the equation for y'_i ,

$$y'_i = \frac{h_{21}x_i + h_{22}y_i + h_{23}}{h_{31}x_i + h_{32}y_i + h_{33}}. \quad (78)$$

Cross multiplying and rearranging gives

$$\boxed{h_{21}x_i + h_{22}y_i + h_{23} - h_{31}x_iy'_i - h_{32}y_iy'_i - h_{33}y'_i = 0} \quad (79)$$

Thus, every point correspondence produces two linear equations in the unknown elements of H .

12 Matrix Form of the Homography Equations

Define the vector containing the unknown homography parameters as

$$\mathbf{h} = [h_{11} \ h_{12} \ h_{13} \ h_{21} \ h_{22} \ h_{23} \ h_{31} \ h_{32} \ h_{33}]^T \quad (80)$$

For a single point correspondence, the two equations can be written as

$$\begin{bmatrix} x_i & y_i & 1 & 0 & 0 & 0 & -x_ix'_i & -y_ix'_i & -x'_i \end{bmatrix} \mathbf{h} = 0 \quad (81)$$

and

$$\begin{bmatrix} 0 & 0 & 0 & x_i & y_i & 1 & -x_iy'_i & -y_iy'_i & -y'_i \end{bmatrix} \mathbf{h} = 0. \quad (82)$$

13 Four Correspondences

Suppose the four correspondences are

$$(x_1, y_1) \longrightarrow (x'_1, y'_1), \quad (83)$$

$$(x_2, y_2) \longrightarrow (x'_2, y'_2), \quad (84)$$

$$(x_3, y_3) \longrightarrow (x'_3, y'_3), \quad (85)$$

$$(x_4, y_4) \longrightarrow (x'_4, y'_4). \quad (86)$$

Stacking the eight equations gives

$$\boxed{A\mathbf{h} = 0} \quad (87)$$

where A is an 8×9 matrix:

$$A = \begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x_1x'_1 & -y_1x'_1 & -x'_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -x_1y'_1 & -y_1y'_1 & -y'_1 \\ x_2 & y_2 & 1 & 0 & 0 & 0 & -x_2x'_2 & -y_2x'_2 & -x'_2 \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -x_2y'_2 & -y_2y'_2 & -y'_2 \\ x_3 & y_3 & 1 & 0 & 0 & 0 & -x_3x'_3 & -y_3x'_3 & -x'_3 \\ 0 & 0 & 0 & x_3 & y_3 & 1 & -x_3y'_3 & -y_3y'_3 & -y'_3 \\ x_4 & y_4 & 1 & 0 & 0 & 0 & -x_4x'_4 & -y_4x'_4 & -x'_4 \\ 0 & 0 & 0 & x_4 & y_4 & 1 & -x_4y'_4 & -y_4y'_4 & -y'_4 \end{bmatrix}. \quad (88)$$

The problem is therefore reduced to finding a non-zero vector $\mathbf{h}$ satisfying

$$A\mathbf{h} = 0. \quad (89)$$

14 Solving the Homography Using SVD

The homogeneous system

$$A\mathbf{h} = 0 \quad (90)$$

can be solved using Singular Value Decomposition (SVD).

Compute

$$A = U\Sigma V^T. \quad (91)$$

The desired solution is the right singular vector corresponding to the smallest singular value. If the singular values are ordered as

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_9, \quad (92)$$

then

$$\boxed{\mathbf{h} = V_{:,9}} \quad (93)$$

where $V_{:,9}$ denotes the last column of V .

The vector $\mathbf{h}$ is then reshaped into

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad (94)$$

Since the homography is defined only up to scale, it is common to normalize it by setting

$$h_{33} = 1. \quad (95)$$

Thus,

$$H \leftarrow \frac{H}{h_{33}}. \quad (96)$$

This procedure is commonly referred to as the **Direct Linear Transform (DLT)**.

15 Affine Transformation as a Special Case

The general projective transformation is

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}. \quad (97)$$

An affine transformation has

$$A = \begin{bmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{bmatrix}. \quad (98)$$

Therefore, an affine transformation is a special case of a projective transformation where

$$\boxed{h_{31} = h_{32} = 0} \quad (99)$$

and, after normalization,

$$\boxed{h_{33} = 1.} \quad (100)$$

Thus,

$$\boxed{\text{Affine transformation} \subset \text{Projective transformation}} \quad (101)$$

16 Affine vs. Projective Transformation

	Affine	Projective
Matrix	3×3	3×3
Degrees of freedom	6	8
Constraints per point	2	2
Minimum correspondences	3	4
Parallel lines	Preserved	Not necessarily preserved
Straight lines	Preserved	Preserved
Perspective effects	Cannot model general perspective	Can model perspective
Typical estimation	Linear system	DLT + SVD

17 Summary

The key ideas are:

1. Homogeneous coordinates allow 2D geometric transformations to be represented using matrix multiplication.
2. Translation is represented by adding the translation parameters in the last column of the homogeneous matrix.
3. Scaling follows directly from $x' = s_x x$ and $y' = s_y y$.
4. The rotation matrix follows from the trigonometric angle-addition identities.
5. An affine transformation contains six degrees of freedom.
6. Each point correspondence provides two constraints.
7. Therefore, three non-collinear point correspondences are sufficient to determine an affine transformation.
8. A projective transformation has eight degrees of freedom because a 3×3 homography is defined only up to scale.
9. Therefore, four point correspondences are sufficient to determine a projective transformation.
10. The homography can be estimated by constructing $A\mathbf{h} = 0$ and solving it using SVD.
11. The DLT method provides a direct linear method for estimating the homography from point correspondences.